WTI Crude Oil Price Prediction using Machine Learning

A COMPARATIVE STUDY OF REGRESSION MODELS FOR COMPLEX FINANCIAL TIME SERIES

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Project Overview

A comprehensive machine learning solution for predicting WTI crude oil prices using multiple regression models and advanced feature engineering techniques.

Objective

Develop a robust prediction system for WTI crude oil prices using historical commodity data and market indicators to forecast future price movements with high accuracy.

Dataset

Daily market data from 2015-2024 including WTI crude oil, heating oil, gasoline, natural gas, stock indices (Dow Jones, NASDAQ, S&P), and commodity prices (gold, silver, copper).

Methodology

Implemented 5 state of the ML algorithms with comprehensive feature engineering, hyperparameter tuning, and rigorous performance evaluation.

Data Processing Pipeline

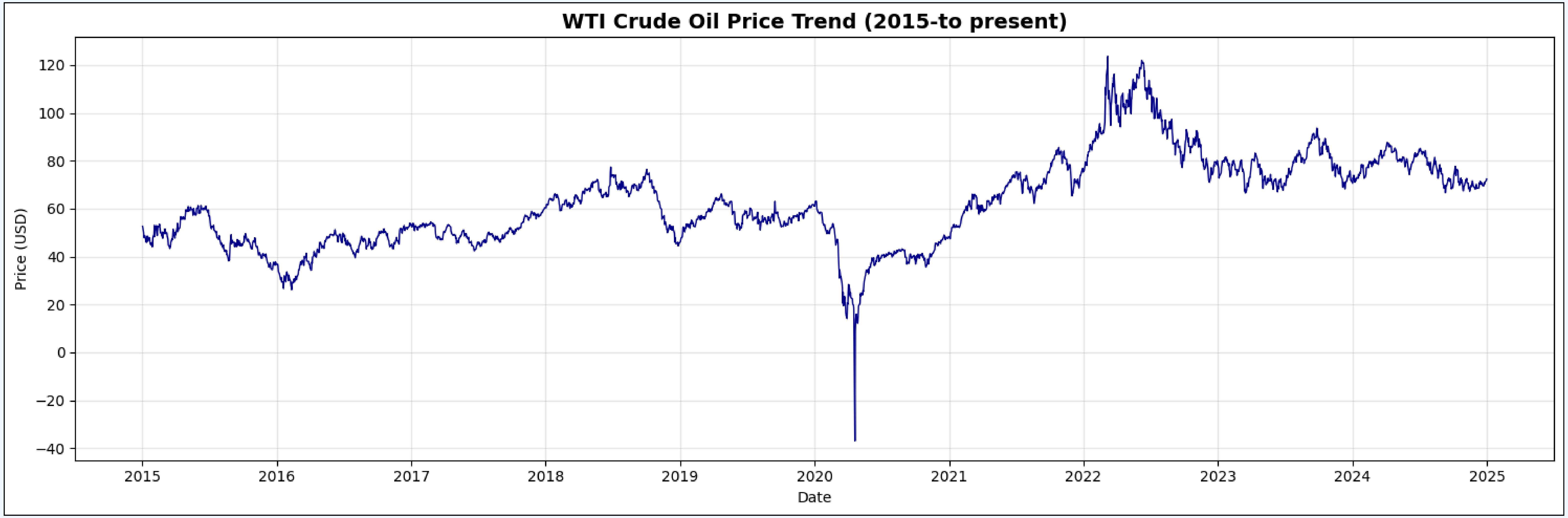
Data Preprocessing

- Handled missing values and data formatting
- Parsed dates and converted numeric columns
- Removed commas from formatted numbers
- Normalized and scaled features

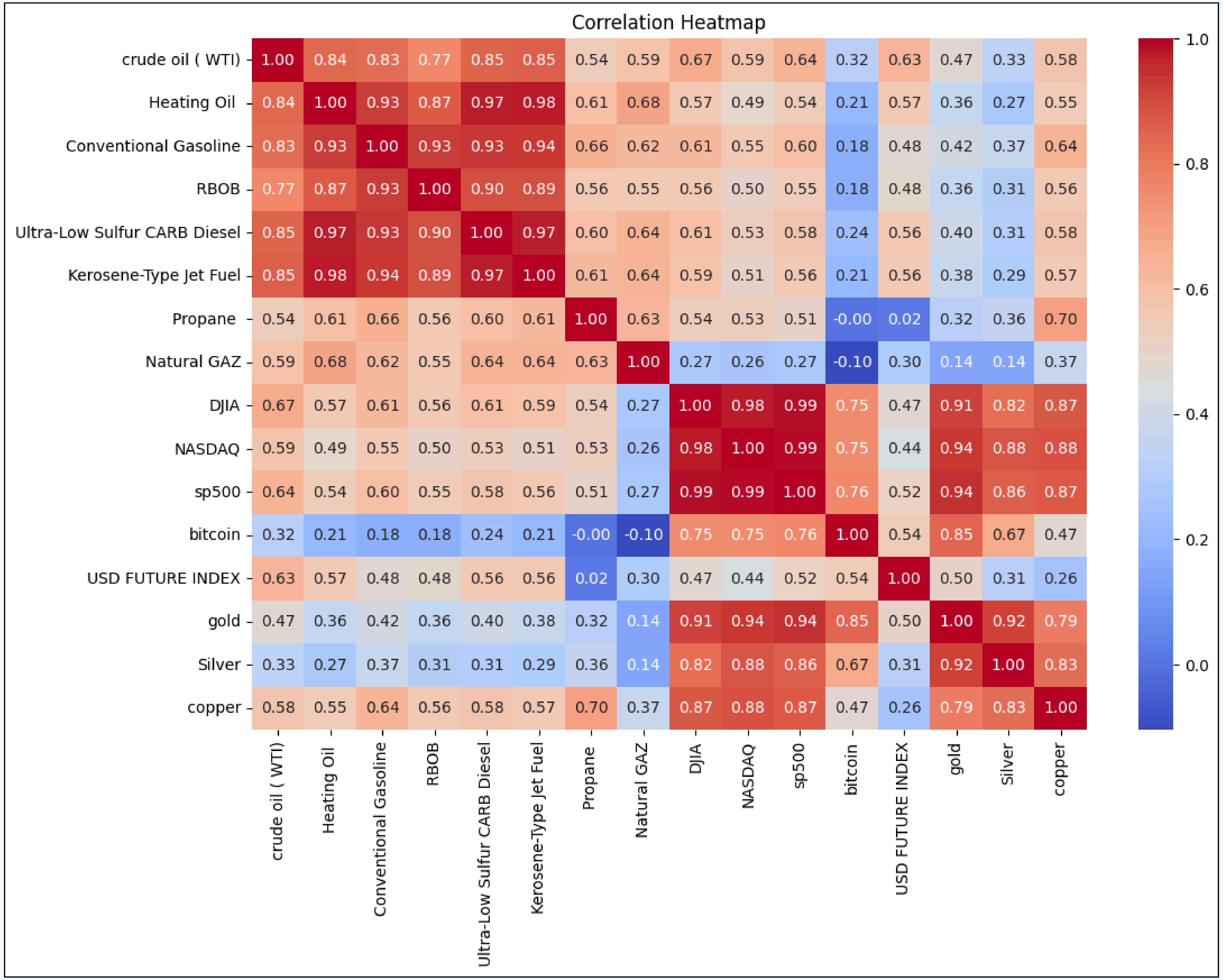
Feature Engineering

- Created lag features (1,3,7,14) days
- Computed rolling statistics
- Generated moving averages
- Analyzed feature correlations

WTI Price Trend Analysis



Feature Correlation Heatmap



Machine Learning Models

5 Advanced Algorithms Tested :

- **Gradient Boosting Regressor:** Ensemble method with sequential learning
- **XGBoost Regressor:** Optimized gradient boosting framework
- **Random Forest Regressor:** Ensemble of decision trees
- **Linear Regression:** Classical statistical approach
- **Support Vector Regression (SVR):** Kernel-based method

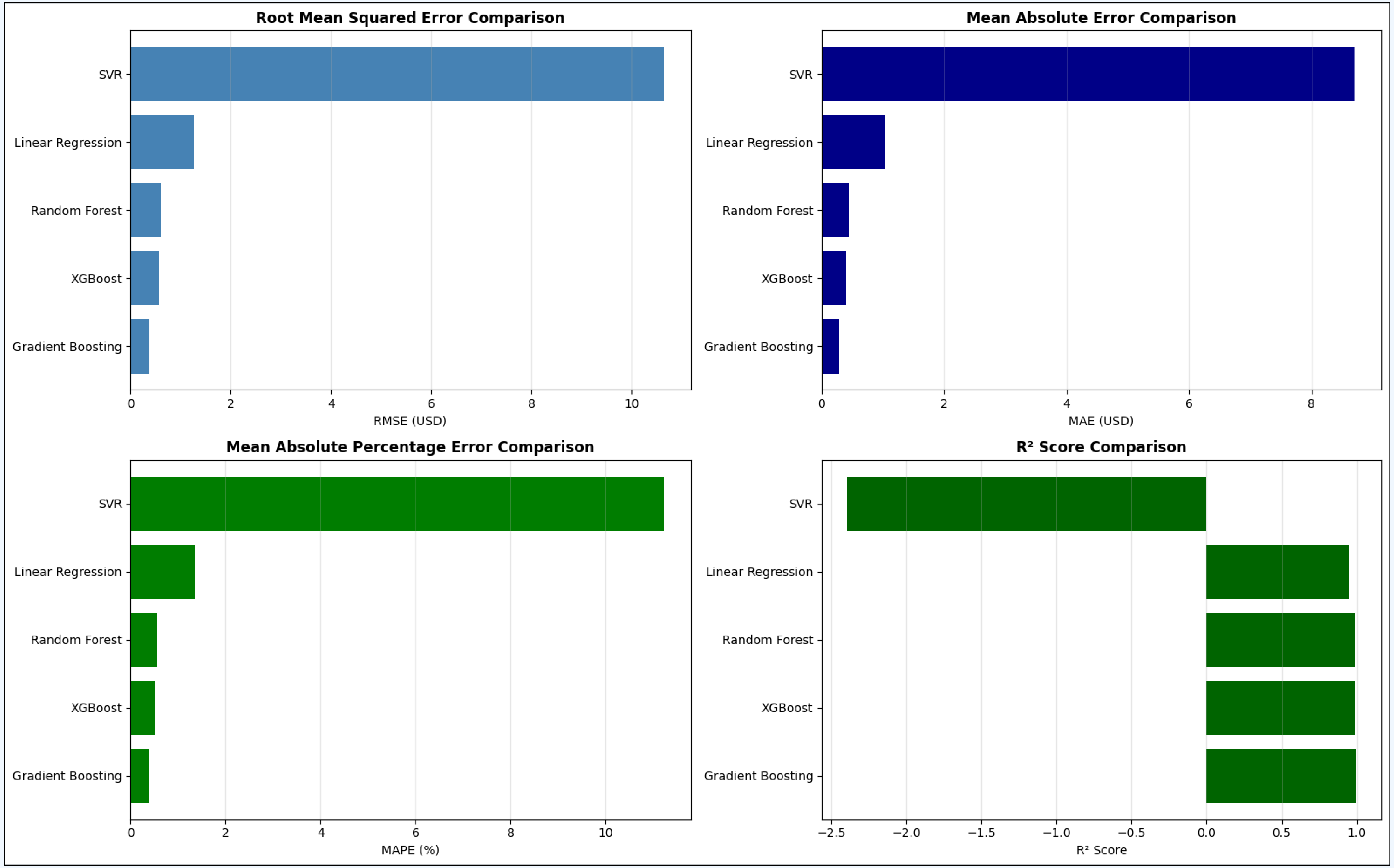
Evaluation Metrics

Evaluation Metrics	Definition
RMSE (RootMeanSquared Error)	Measures the average magnitude of prediction errors in the same units as the target variable. (Lower values are better).
R^2(Coefficient of Determination)	Measures the proportion of variance in the target variable explained by the model. Ranges from 0 to 1 (or negative for terrible models). (Closer to 1.0 is better).
MAPE (MeanAbsolute Percentage Error)	Measures the average error as a percentage of actual values. Useful for comparing models across different scales.(Lower percentages are better).

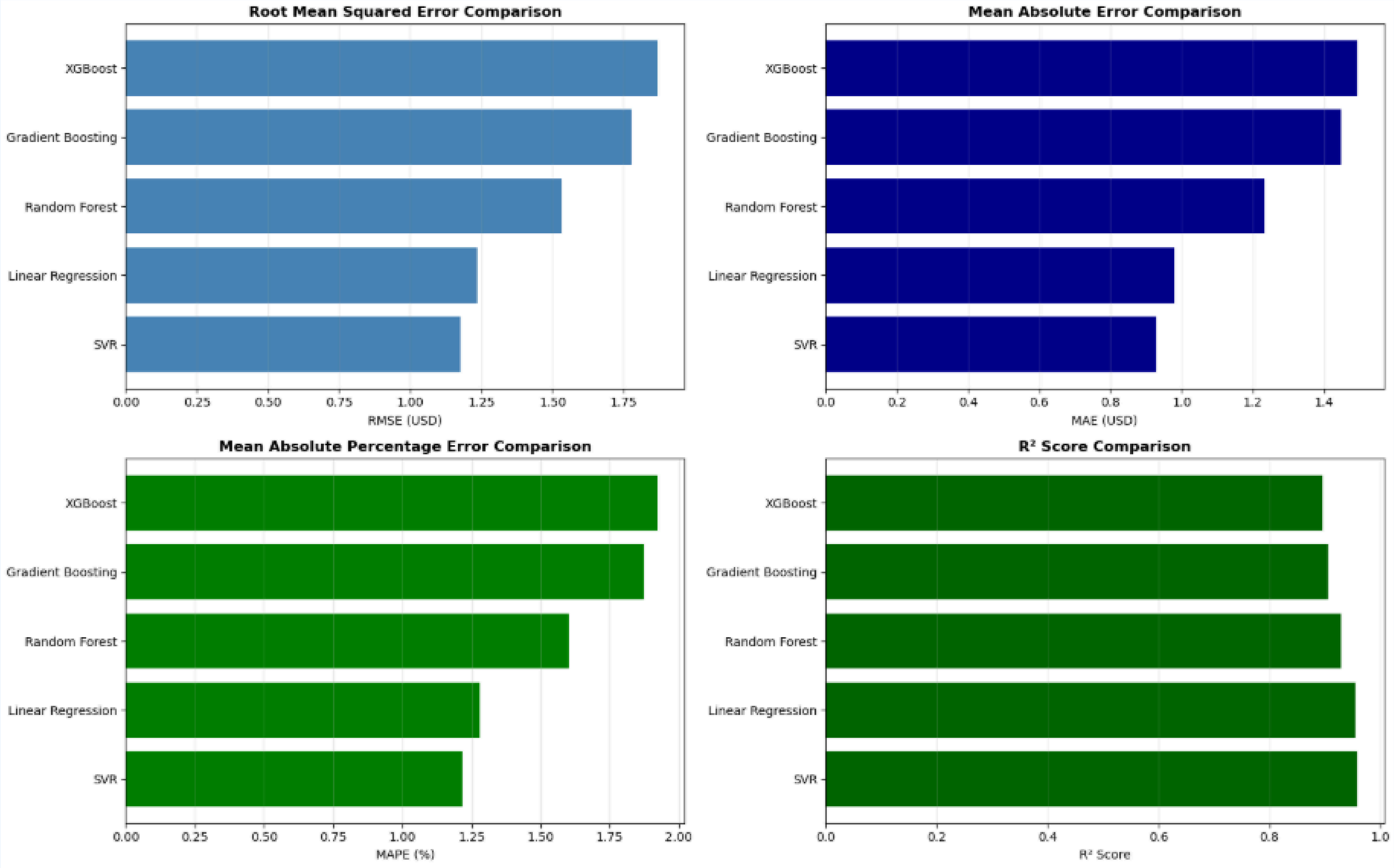
Model Performance Comparison

Model	RMSE	MAPE (%)	R^2 Score	MAE
🏆SVR	1.175	1.216%	95.85%	0.928
Linear Regression	1.234	1.279%	95.42%	0.979
Random Forest	1.533	1.602%	92.94%	1.231
Gradient Boosting	1.778	1.868%	90.51%	1.448
XGBoost	1.824	1.895%	90.01%	1.493

Model Performance Comparison



Model Performance Comparison 2



Winner: Support Vector Regressor

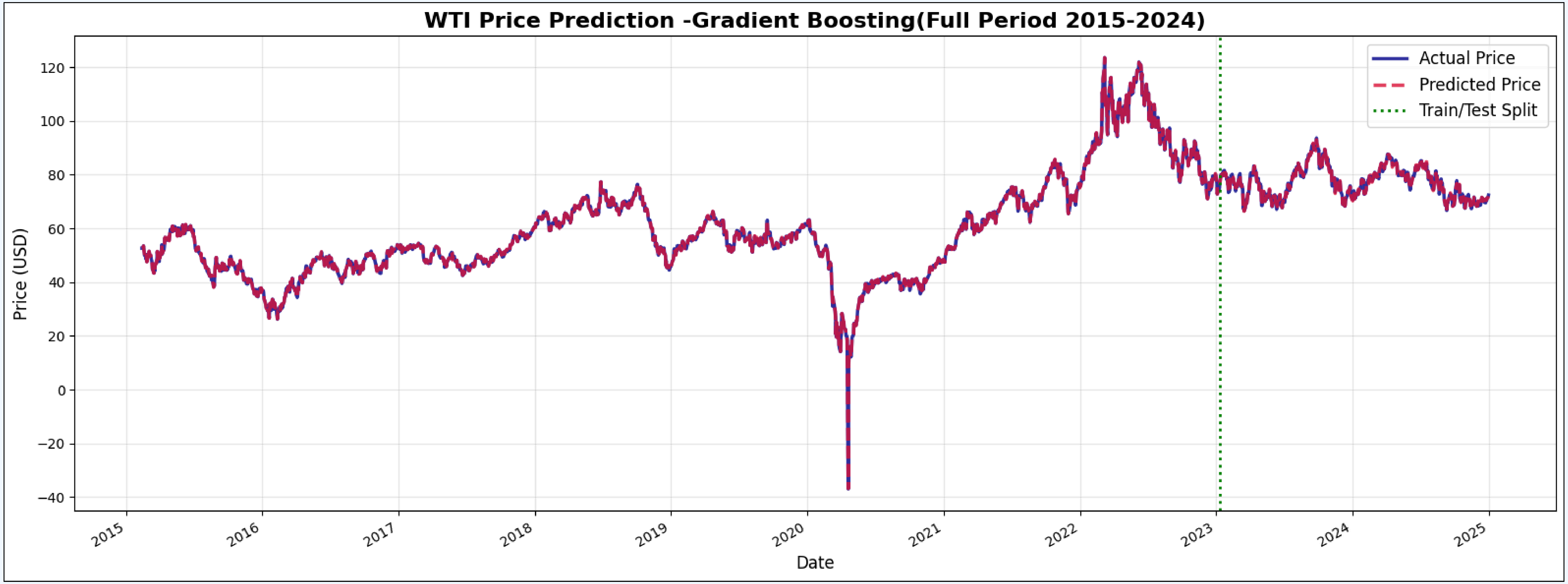
Why It Won ?

- Lowest error rates (MAE: 0.928)
- Highest accuracy (R^2: 95.85%)
- Captures data patterns effectively
- Linear kernel with optimized C parameter provides excellent precision

Key Insights

- All models are reliable
- Ensemble methods outperformed
- Tree-based models captured complexity
- Feature engineering was crucial
- Model ready for production

Best Model: Predictions vs Actual Data



Streamlit Web Application

- **Dataset & Target**
Overview of the data we're working on

- **Interactive Manual Prediction**
Adjust feature values via sliders for instant WTI price predictions

- **Model Performance**
Compare models metrics and identify the best model

- **Forecasting**
Predicting WTI prices for the next 7 days using the best performing model

- **Prediction Visualization**
Compare predicted WTI prices from different models on the same plot to see trend differences.

Conclusions & Future Work

Key Achievements

- Achieved 95.85% R^2 accuracy
- Comprehensive model comparison
- Production-ready solution
- Robust feature engineering

Future Improvements

- Deep learning models (LSTM)
- Real-time prediction API
- Additional economic indicators
- Multi-step forecasting

Thank You

For Your Attention

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